Molar Concentration

x M of Ex in y ml of water:

$$y\,ml\ H_2O \left(\frac{x\ moles\ Ex}{1000\ ml\ H_2O}\right)\left(\frac{mass\ molar\ of\ Ex}{1\ mol\ Ex}\right)\left(\frac{\forall\ Ex}{mass\ Ex}\right)\left(\frac{\forall\ Ex}{\forall\ Ex * \%}\right) = \boxed{z\ ml\ Ex}$$

Molar concentration — mass molar — volumetric density — purity

Example: 0.1 M H_2SO_4; H_2SO_4 to 98.5 %

Mass molar:

$$\begin{array}{rl} & 2 * 1.008\,g\ H \\ + & 1 * 32.06\,g\ S \\ + & 4 * 16\,g\ O \\ \hline & 98.076\,g\ H_2SO_4 = 1\ mol\ H_2SO_4 \end{array}$$

$$250\ ml\ H_2O \left(\frac{0.1\ mol\ H_2SO_4}{1000\ ml\ H_2O}\right)\left(\frac{98.076\,g\ H_2SO_4}{1\ mol\ H_2SO_4}\right)\left(\frac{1\ ml\ H_2SO_4}{1.83\,g\ H_2SO_4}\right)\left(\frac{100\ ml\ H_2SO_4}{98.5\ ml\ H_2SO_4}\right) = \boxed{1.36\ ml\ H_2SO_4}$$